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"Ant-Wall" model to study drug release from excipient matrix

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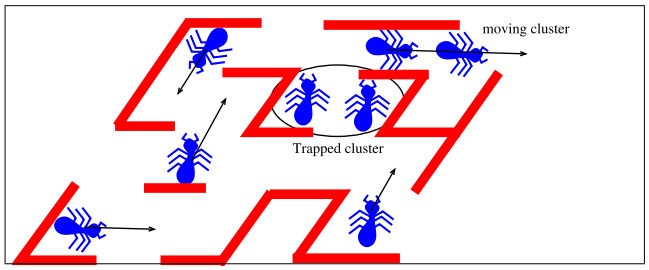
HIGHLIGHTS

- Controlled drug release through excipient matrices is studied using a lattice model.
- Drug molecules modeled as blind ants (hydrophilic) and friendly ants (hydrophobic).
- Excipients modeled as walls of matrix that the ants can cross with a probability.
- Captures the effect of matrix and drug properties and excipient–drug interactions.
- Optimum drug loading is near the percolation threshold for hydrophobic drugs.

GRAPHICAL ABSTRACT



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ABSTRACT

We propose a two-dimensional lattice model to study the effects of drug loading, matrix structure, and drug–excipient interactions on the drug release through excipient matrix. Analogy is made with the random walk of ants (drug molecules) in a maze (excipient matrix). Ants can be "blind" (non-aggregating) or "friendly" (aggregating), representing hydrophilic and hydrophobic drugs, respectively. Excipient–drug interactions are accounted as the probability of ants (drug molecules) crossing walls (excipient molecules). Monte Carlo simulations of the model are performed to obtain the amount of drug escape (release) as a function of time for different values of drug loading, excipient–drug interaction, and matrix size. Although a Weibull function is able to fit the drug release in the entire time range for the hydrophilic case, two Weibull functions are required in the hydrophobic case signifying an initial burst release followed by a long-time sustained release. The competition between the drug escape and drug aggregation results in the existence of an optimum drug loading for sustained drug release in the case of hydrophobic drugs. Also, the drug release is best controlled at moderate excipient–drug interactions, since strong excipient–drug interactions results in substantial amount of trapped drugs that are never released. Results of this study may provide design rules of matrix formulations for the delivery of poorly soluble drugs and stimuli-responsive drug delivery formulations.

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1. Introduction

Controlled release, oral drug formulations often consist of drugs loaded within an excipient matrix, which results in a dramatic decrease in drug diffusivity and subsequent reduction in drug release rates, compared to pure drug formulations without excipients. These can specially be useful for the delivery of drugs with low aqueous solubility [1,2], which tend to form aggregates/crystals after dissolution thus resulting in poor intestinal absorption. Poorly soluble drugs can be loaded at supersaturated concentrations in an excipient matrix, as slower drug release rates would inhibit the tendency of aggregation in the gastro-intestinal (GI) tract. Yet another application of excipient matrix is in sustained release applications [3] to minimize dosage frequency, since drug concentrations can be maintained within the therapeutic window for a reasonably long period of time. Although the role played by the excipient matrix is well understood at a qualitative level, actual design of such formulations are far from trivial. Apart from the essential requirements of bio-compatibility and safety, three important questions need to be addressed to be able to develop the design rules for the matrix formulations:

1. How much drug loading (excipient-to-drug ratio) is optimal?
2. What is the effect of matrix structure and size?
3. How to account for the drug/excipient hydrophobicity and drug–excipient interactions?

In general, the answers to the above questions demand a series of experiments for a range of drug loading, matrix structures, and drug/excipient combinations. Required experimentation may be substantially reduced by the use of theoretical/simulation approaches [4–8] that can provide a conceptual understanding of drug release from matrices.

Theoretical and numerical studies of controlled drug release can be classified into three categories; (1) empirical and semi empirical models like power law (Peppas model) [9], Weibull function [10–12], Hopfenberg model [13,14]; (2) mechanistic models based on Fick's law of diffusion [4,5,15,16], and (3) lattice based stochastic models [17–29]. Weibull function [10–12] is one of the most used empirical model for the interpretation of drug release data; in fact, Higuchi law and the power law model can be considered as limiting cases of the Weibull function.[20] Despite their widespread use, empirical and semi-empirical models do not explicitly include the effects of matrix structure and drug–excipient interactions, and thus have limited applicability in developing design rules for matrix formulations. Mechanistic models based on the solution of Fick's law are advantageous in a theoretical sense, but usually involve an assumption regarding the matrix structure [4,5,30], e.g., uniform initial drug distribution inside the matrix or uniform matrix density. Moreover, analytical solutions of such models are only available for simplified geometries and boundary conditions; numerical solutions are required for complicated geometries or in the case of position-dependent diffusion coefficients. These discrepancies result in subtle deviations from the model predictions in many cases, often termed as non-Fickian behavior. Although simulations of lattice models also follow diffusion theories, they do explicitly include the matrix structure and thus can be considered significant improvement over the mechanistic models. However, the choices of lattice representation and lattice dimension, and inclusion/exclusion of various energetic interactions in simulations are non-trivial issues and requires some consideration. Besides matrix structure, matrix size also plays a crucial role in drug delivery. For homogeneously distributed drugs in microspheres, the release duration typically increases with increase in particle size but the drug distribution is also found to depend on the method of synthesis [31,32]. Here, we are interested in such drug delivery systems where both matrix size and matrix structure plays an important role in drug release behavior.

The basic scheme of lattice simulations [17,18] involve drug molecules performing random walk on a lattice. "Drug release" is defined as either the escape through a pore created on the lattice [18] or from the ends/surface of the lattice [17,20,27,28]. Fractal nature of the matrix is sometimes incorporated by confining drug release within a percolated fractal [17,19,21]. Interaction between drug molecules are typically included by imposing limitation on the number of molecules that can occupy a lattice site [18] and drug–excipient interactions are often included by imposing a constraint that drug molecules cannot pass through the fractal boundaries. Further, regions of different diffusivity inside the matrix can be modeled by varying the probability of movement between different regions [21,28]. Lattice simulations have been performed on a variety of matrix geometries, structures, and lattice dimensions. However, certain important features are lacking in the existing studies, which provide scope for further improvement. First, hydrophobic drug molecules can form aggregates or crystals and then move in a concerted manner instead of moving individually. Second, the drug molecules should be allowed to pass through (break up from) excipient molecules with a probability (p) that is a function of drug–excipient interaction; only $p = 0$ (fully impermeable barrier(s)) and $p = 1$ (no barrier) cases have been studied so far. This is specially relevant in the case of stimuli-responsive release, since the magnitude of p can change with matrix swelling or protonation levels of drug/excipients. Third, studies have been performed either in free-diffusion (no excipient) or percolated case. Systematic studies with the change in matrix structure by variations in the amount of excipient have not been performed. In this paper, we propose a simulation method that address these three aspects.

For simplicity's sake, in our model, we will use the analogy with ants performing random walk on a maze, where the ants and maze represent the drug and excipient molecule, respectively. Such analogies have been used earlier in the context of controlled release [24,33] and more generally, in the description of percolation phenomena [34,35]. We solve our model for two distinct cases, referred as "blind ant" (analogy with hydrophilic drugs) and "friendly ant"(analogy with hydrophobic drugs). Blind ant [36] attempts to move in all possible directions with equal probability, but the moves resulting in an overlap of ants are rejected. Friendly ant, introduced in this work, behave as a blind ant except when they are part of an ant cluster (aggregate), in which case the cluster (as a whole) attempts to move in all possible directions with equal probability. In

both these cases, moves are accepted only if the new position is not already occupied and with a probability p for each wall crossing. In the following section, we first describe the "ant-wall" model and simulation methodology, followed by the physical significance of model parameters in the context of drug delivery. Results of simulations of drug release through excipient matrix is discussed afterwards. It is worth pointing out that the models developed here can be adopted to several other problems in diverse disciplines, especially those involving molecules in a crowded environment.

2. Methodology

Fig. 1 shows the basic scheme of our "ant-wall" model, which consists of a 2D $N \times N$ lattice occupied by ants and walls. Ants can occupy any of the N^2 cells (sites) of the lattice, with no more than one ant per site. Walls can occupy any of the $2N(N+1)$ cell boundaries, with a maximum of one wall per cell boundary. Cell boundaries can be classified into "surface" (ends of the lattice) and "bulk" (inside the lattice) boundaries. a and w are the ant fraction (number of ants divided by N^2) and wall fraction (number of walls divided by $2N(N+1)$), respectively. p is the probability of an ant crossing a wall, given that the attempted move is in the direction of the wall and does not lead to a cell already occupied by an ant. It is useful to look at the extreme cases of the parameters (N, a, w, p). $a = 0$ and $a = 1$ represent empty and fully occupied matrix, respectively. Similarly, $w = 0$ and $w = 1$ represent fully open (no walls) and fully closed matrix (all possible walls), respectively. $p = 0$ and $p = 1$ indicate fully impermeable and fully permeable wall, respectively. "Neighbors" of ants are defined as ants present in adjacent locations on the top, bottom, left, and right, with no common wall between them; diagonal locations are excluded. We solve the model for two cases: blind ant (Fig. 1A and B) and friendly ant (Fig. 1C and D) as discussed earlier. In the friendly ant case, an ant is considered part of an "ant aggregate" if at least one of the ant in the aggregate is its neighbor. An ant with walls on all its sides is referred as a "trapped ant" in the $p = 0$ case.

It is worth pointing out that the surface effects diminish at large N; $N \to \infty$ represents ants diffusing in a bulk random media. To characterize our system in this limit ($N \to \infty$), we did some preliminary simulations using periodic boundary condition in both horizontal and vertical directions. Simulations were performed for the case of "ideal ants" (two or more ants can occupy same site) and non overlapping ants. We observed that in the absence of walls, motion of both ideal ants and non-overlapping ants is diffusive but it becomes sub-diffusive in the presence of walls [37–39] (for details, see supplementary information). Since the focus of this study is on the drug release from finite-size systems, we perform all the other simulations reported in the paper without the periodic boundary conditions, i.e., ants may escape the lattice from the surface.

We begin with a random distribution of ants and walls on cells and cell boundaries, respectively. In a Monte Carlo (MC) move, we randomly pick one ant (blind ant case) or an ant aggregate (friendly ant case) and choose any one out of the four possible directions with equal probability. The move is always accepted if there are no wall crossings and the chosen direction points to cell(s) that are not occupied by other ant (ant aggregate). If the move involves wall crossings, each crossing is accepted with probability p, provided the new position(s) is not already occupied. Since all such wall crossings must be accepted to result in a movement of an aggregate, the actual probability of accepting an aggregate move is p^c, where c is the total number of wall crossings. Ant is considered to "escape" the lattice if the chosen direction results in a movement outside the lattice; escaped ant is considered lost (number of ants reduced by one) and cannot reenter the lattice. The remaining ants in the aggregate from which an ant has escaped are still assumed to move as an aggregate. Again, if there is no wall in the direction of escape, the move is always accepted. If, however, there is a wall in that direction, the move is accepted with a probability p for each wall crossing (p^c for friendly ant case). It is important to point out that in the friendly ant case, only those ants in the aggregate that leave the lattice are considered to have escaped. One MC step refers to number of MC moves equal to the number of ants in the blind ant case, and the number of ant aggregates in the friendly ant case, where single ants that are not part of any aggregate are also counted as an aggregate containing one ant. The rationale behind this is that we assume that there is no inner diffusion of ants within an aggregate and the aggregate moves as a whole. Although the MC step can be related to a time duration inversely proportional to the diffusivity, as routinely done in kinetic Monte Carlo simulations [40,41], we adopt a simplified description in this work and study the release (escape) behavior as a function of the number of MC steps, t_{MC}. t_{MC} needs to be multiplied by a factor proportional to σ^2/D to get the actual time, where σ and D are the cell size and ant diffusivity, respectively. Please note that as discussed in an earlier study [41], the proper treatment of the size-dependence of diffusivity of aggregate will require the incorporation of hydrodynamic interactions, which are not considered in our simulations. For a given set of model parameters (N, a, w, p), we run each simulation for $t_{MC} = 10{,}000$ and the results are averaged over 1,000 such simulations (starting from a new randomly generated lattice) to calculate the average number of escaped ants as a function of t_{MC}. Simulations are performed using a C++ code with the random number generators borrowed from the GNU Scientific Library (GSL) version 2.3.

Analogy between the ant-wall simulations and drug release from excipient matrix can be understood in the following way. The 2D lattice is a representation of matrix geometry, which can be easily generalized to other shapes and lattice coordination numbers (maximum number of nearest neighbors). N multiplied by the cell size σ is the size of the matrix, σ is also representative of the molecular size of drug molecules. aN^2 is the number of drug molecules and w/a is a measure of drug loading (excipient to drug ratio). p is a measure of excipient–drug interaction, with higher values of p characterizing weaker interactions. Loosely speaking, $p \propto \exp(-\Delta E/k_BT)$, where ΔE is the excipient–drug binding energy, k_B is the Boltzmann constant, and T is the temperature (in Kelvin). Strong interactions such as covalent or ionic bonds ($\sim 100k_BT$) between the drug and excipient molecules will therefore result in extremely small values of p and $p = 0$ is a good approximation in such cases. Nonzero p values are required in the case of moderate interactions, such as hydrogen bonding between donor

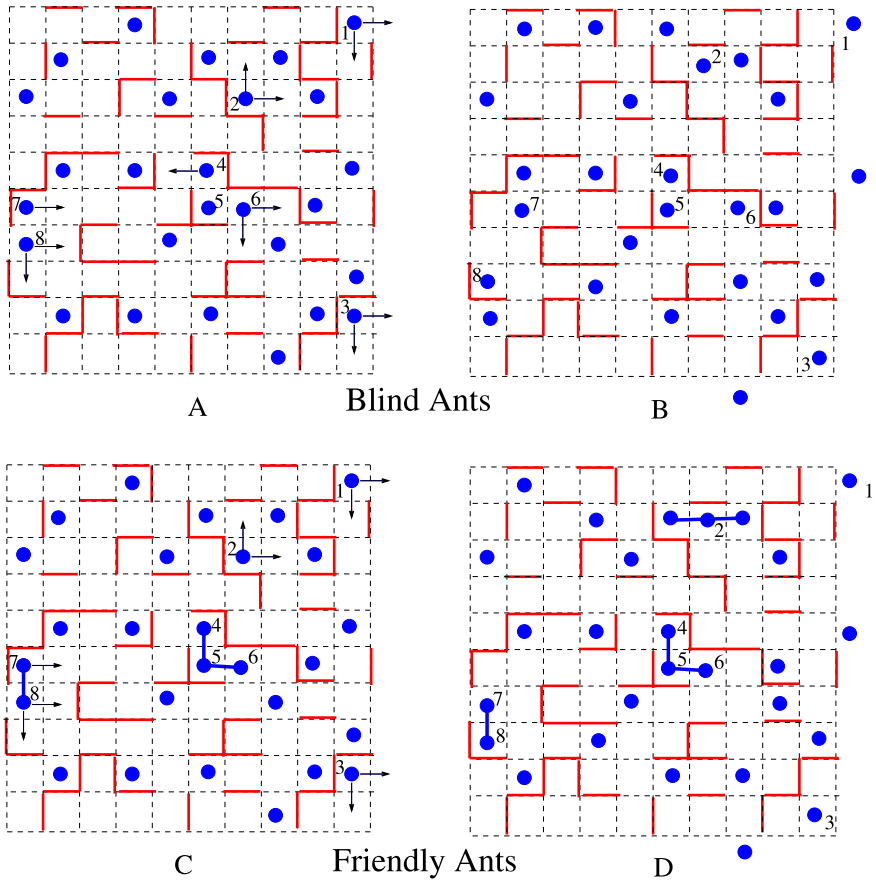

Fig. 1. Ant-Wall Model. Configurations of ants (blue circles) inside the matrix with walls (red bond) at two consecutive time steps for blind ant (A and B) and friendly ant (C and D) models. Blue bond between two ants in C and D represents that ants are part of same cluster. We have numbered few ants in the matrix to show their motion. In (A) and (C), arrows represents all possible moves of the numbered ants. Ants 1, 2 and 3 has two possible moves for both the models. Ants 4, 5 and 6 forms a cluster with no possible move in friendly ant case; these are trapped ants, which will never escape the matrix if $p = 0.0$. In blind ant case, Ants 4 and 6 have one and two possible moves, respectively. Ants 7, 8 form another cluster in the friendly ant case and move together with only two possible moves, whereas in blind ant case, Ant 7, 8 moves are independent. (B) and (D) show the configuration of ants after ants have moved. Ant 1 escapes out of surface boundary and become free. Ant 2 moves in between other two ants and formed a new cluster in the friendly ant case. Ant 3 moves down alone also in the friendly ant case, since there is a wall between it and other ants in adjacent cells. Ant 7, 8 moves down together as a cluster in friendly ant case.

and acceptor groups on the drug and excipient molecules. Actual magnitude of hydrogen bonding energy depends on the properties of the donor and acceptor (e.g., charge) and energies up to 50 k_BT can be observed, but most typical hydrogen bonding energies are in the range of $3 - 5$ k_BT for a donor–acceptor pair, which gives $p = 0.02 - 0.05$.[42] Though the use of p to characterize excipient–drug interactions is rather simplified and do not account for factors such as the range of these interactions, it succeeds in capturing three types of drug-release behavior obtained for high drug loading:

- Strong interactions ($p \approx 0$): Incomplete (partial) drug release.
- Moderate interactions ($0 < p \ll 1$): Sustained drug release.
- Weak interactions ($p \approx 1$): Excipient does not control drug release.

Further, simulations of a given system for different values of p can also be thought to characterize the stimuli-responsive behavior of the matrix. For example, hydrophilic matrices may swell by absorbing water, which enhances drug release by increasing its molecular mobility [43,44]. On the other hand, water absorption will be lower for hydrophobic matrices, thus resulting in little or no swelling and subsequent changes in drug mobility. Since similar control on drug mobility is obtained by changes in p, results of different p should correspond to matrices of different hydrophobicity. It is worth pointing out that since the solvent is not explicitly included in our simulations, hydrophobicity in our discussion refers to effective interactions between the components of the system that in reality are solvent-mediated. However, a proper characterization of such systems also requires the knowledge of the extent of swelling (i.e., change in cell size σ) and change in ΔE with swelling. In addition, p can be varied within the simulations to study the influence of such dynamic changes on the drug release behavior.

3. Results and discussion

Drug release in our lattice model can be considered as a diffusion process with geometrical constraints imposed by the walls. For higher values of both the ant and wall fraction, we observe complementary fractal-like structures of both ants and walls. Ant and wall percolation would occur for ant and wall fractions beyond the bond and site percolation thresholds, which are $w_c = 1/2$ and $a_c \approx 0.593$, respectively. This type of release problems can be modeled as fractal-like reactions that are specified by time dependent reaction rates [45]. As will be discussed later, it is rather convenient to fit the simulation results with the Weibull function in order to understand the physical behavior. The rationale behind the use of Weibull function can be understood using the approach of Kosmidis et al. [20,21], which is discussed briefly below.

Drug escape rate from the matrix will depend on fraction $f(t)$ of the drug molecules on the open surface boundary at time t. Rate of drug escape from the matrix can be effectively written as [20]

$$-\frac{dN}{dt} = \alpha f(t)N, \tag{1}$$

where N and $f(t)N$ are the total number of drug molecules and the number of drug molecules on the surface, respectively at time t, and α is some proportionality constant. Fraction of drug on the surface will depend on a, w, and p and we therefore assume [20] $f(t) = t^{-m}$, where the exponent m depends on a, p and w in our model. Substituting $f(t)$, Eq. (1), becomes

$$\frac{dN}{dt} = -\alpha t^{-m} N. \tag{2}$$

Solving Eq. (2), we get $N = N_0 \exp\left\{(-t/\tau)^{\beta}\right\}$, where $\beta = 1 - m(a, w, p)$, $\tau^{\beta} = 1/\alpha$, and N_0 is the initial number of drug molecules in the matrix. Amount (%) of drug released from the matrix in time t can then be written as the Weibull function,

$$M(t) = M_{\infty}\left[1 - \exp\left\{-\left(\frac{t}{\tau}\right)^{\beta}\right\}\right], \tag{3}$$

where M_{∞} is the total % drug released from the matrix in infinite time (= 100% if there are no trapped drugs). β, which depends on m, gives the information about the shape of release profile. τ is the characteristic time by which ≈63% of M_{∞} is released. Note that the percentage of trapped drug in the system is $100 - M_{\infty}$. As discussed in Kosmidis et al. [20], Eq. (3) reduces to the pure diffusion case for $m = 0$ and its short-time (Taylor series) approximation results in the power law model [9].

We first studied the blind ant case, i.e., the case of hydrophilic drug molecules. Fig. 2A,B,C shows the effect of wall (excipient) fraction on the drug release from the matrix [46] of size $N = 10$ for three different values of p. For $p = 0.0$ (Fig. 2A), Weibull function fits release profiles for all values of w shown in the figure, with M_{∞} taken as the saturation (long-time) value of the drug release obtained from simulation. As w increases, M_{∞} decreases that implies an increase in the amount of trapped drug. This is consistent with the fact that strong interactions between the drug and excipient molecules can lead to trapping of drugs inside the matrix and only a certain % of loaded drug can be released (incomplete release). We also observe dramatic decrease in M_{∞} beyond $w = 0.6$ in Fig. 2A, that is, $M_{\infty} > 90\%$ for $w = 0.0, 0.2, 0.4$, but reduces to ≈70% for $w = 0.6$ and ≈20% for $w = 0.8$. This can be attributed to the fact that the number of closed loops are expected to increase dramatically beyond the bond percolation threshold, i.e., $w > (w_c = 1/2)$. Note that no such trapping occurs for $p > 0$ case (Fig. 2B and C) where $M_{\infty} = 100\%$. However, sustained drug release may occur over longer durations for lower values of p (for $p = 0.02$ in Fig. 2B) than compared to higher p values (for $p = 0.1$ in Fig. 2C), as evidenced by higher τ values in the $p = 0.02$ case.

For $p = 0.02$ in the blind ant case (Fig. 2B), drug release profile fits well with the Weibull function (Eq. (3)) for small to intermediate w values. However, for high values of w (up to $t_{MC} \sim 100$ for $w = 0.8$ in Fig. 2B), short-time drug release behavior show significant deviation from the best fitted Weibull function. This deviation arises from the fact that the highly obstructed path of drug molecules in the $w = 0.8$ case results in a slow escape from the lattice, except for the drug molecules that are at the surface initially. We therefore observe an initial burst release (up to $t_{MC} \lesssim 10$) followed by a slow release regime ($10 \lesssim t_{MC} \lesssim 100$); Weibull function behavior is re-established at longer times ($t_{MC} \gtrsim 100$) after a substantial fraction of drug has been released. Further increasing p (e.g., Fig. 3C for $p = 0.1$) results in a progressively weaker dependence of drug release on w and Weibull function behavior is again observed. This is expected, since the $p = 1$ case for any value of w is identical to the $w = 0$ case (free diffusion). The variation of fitting parameters β and τ with w is shown in Fig. 2D. τ increases monotonically as w increases, consistent with the expectation that drug escape rate should decrease with an increase in the number of walls. Also, τ decreases with an increase in p, as the excipient–drug interactions become weaker. Surprisingly, β shows a non-monotonic dependence on w (inset of Fig. 2D) with an initial decrease followed by an increase with increasing w. Therefore, the function $f(t) = t^{-m}$ appearing in Eq. (1) with $m = 1 - \beta$ first increases then decreases with increase in w. Moreover, β minima (also $f(t)$ minima) for all p is found at $w = 0.6$, i.e., after the bond percolation threshold $w_c = 1/2$. A tentative explanation to these observations is that the drug molecules at the surface of the lattice may diffuse inside the lattice instead of diffusing outside (escape), the magnitude of inside diffusion become very small when the fraction of walls increases beyond $w_c = 0.5$.

Next, we studied the friendly ant case, i.e., the case of hydrophobic ants (Fig. 3) for $N = 10$ matrix. For $p = 0$ case in Fig. 3A (and $w = 0$ case in Fig. 3B,C), the behavior obtained for hydrophobic drugs is qualitatively similar to the hydrophilic drugs

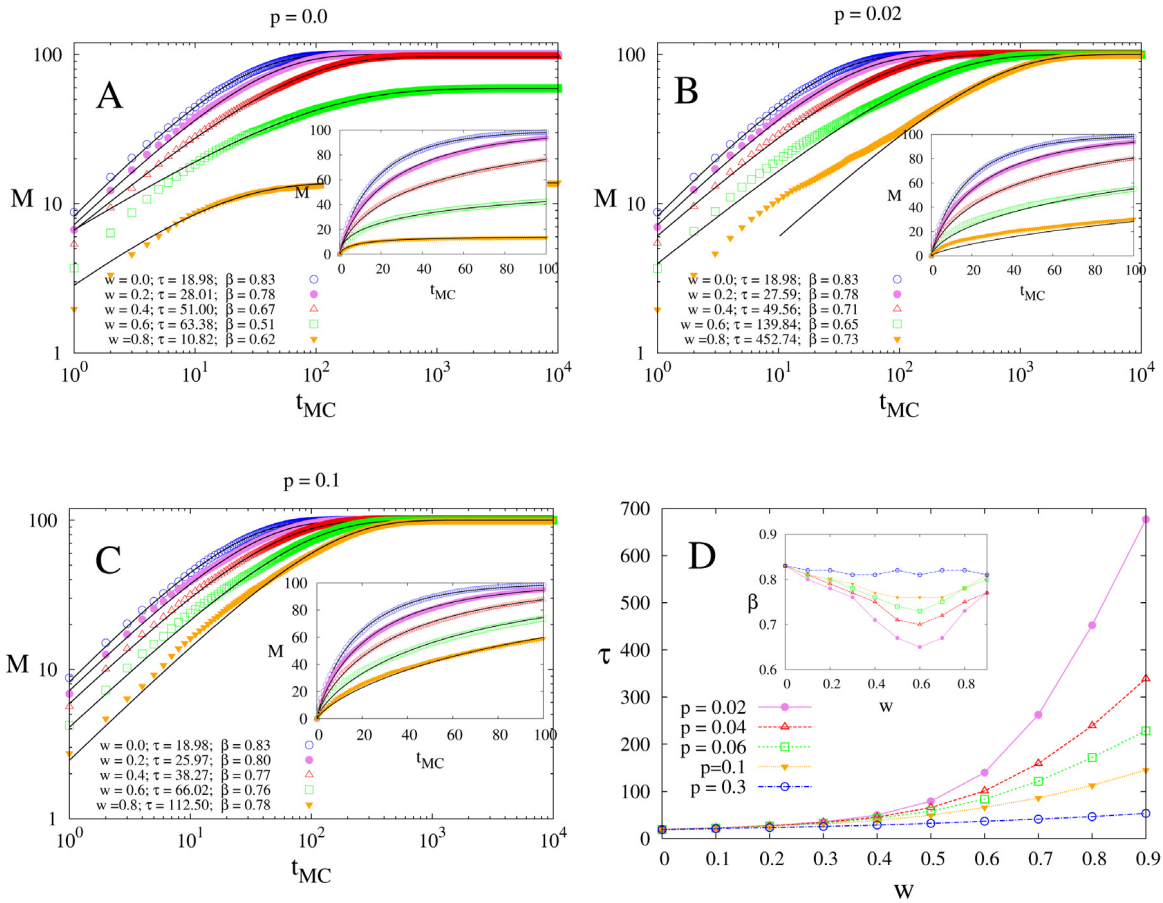

Fig. 2. Blind ant simulations (hydrophilic drugs). (A), (B), (C) shows the log–log plot of M vs t_{MC} for $p = 0.0$, $p = 0.02$, $p = 0.1$, respectively, for $N = 10$, $a = 0.2$, and different values of w. Insets in (A), (B), (C) shows corresponding drug release profiles till $t_{MC} = 100$. Solid lines indicates Weibull function fits with parameters β and τ mentioned in the legends. In (A), M_∞ is different for different w, but in (B) and (C), all plot saturates at 100%. (D) shows dependence of τ on w for different p. β vs w is plotted for different values of p in the inset of (D).

(Fig. 2A) and fits well with the Weibull function. However, drastic differences between the hydrophobic and hydrophilic cases are observed for $p > 0$ (compare Fig. 3B,C with Fig. 2B,C). First, unlike the hydrophilic case, M_∞ is no longer 100% for $p > 0$. Second, long-time release behavior is very slow in the hydrophobic case, resulting in much higher τ and lower β values when compared to hydrophilic case. In fact, short-time release behavior can also be fitted with another Weibull function with higher β and lower τ values (referred as β_i and τ_i in Fig. 3), that is, a mixed Weibull type of release behavior is observed. Third, while $M(t)$ decreases with increasing w for the $p = 0$ case (Fig. 3A) similar to that observed in the hydrophilic case for all values of p (Fig. 2A,B,C), a nonmonotonic increase is observed for $p \neq 0$ in the hydrophobic case (Fig. 3B,C) beyond $t_{MC} \gtrsim 100$. Fourth, τ versus w plots (Fig. 3D) show the presence of a maxima around $w \approx (w_c = 1/2)$ that correspond to an optimal value of w for sustained release. Since τ in Weibull function corresponds to $\sim 63\%$ drug release; it is calculated from long time fit $t_{MC} > 10$. These effects can be understood by noting that in addition to the drug escape from the lattice, drug molecules aggregate with a kinetics $A_n + A_m \rightarrow A_{n+m}$, where A_n, A_m and A_{n+m} are the aggregates having n, m and $n + m$ drug molecules respectively. In the beginning, we observe the escape of free drug molecules that are not part of an aggregate, giving rise to the first Weibull function. Second Weibull function corresponds to the escape of drug molecules in the aggregates, which is a much slower process as the aggregates diffuse much slowly, especially in the presence of walls. Naturally, the drug release rate decreases with increasing w in the first Weibull regime, but it may increase with increase in w in the second Weibull regime as aggregation rate decreases with increasing w.

We have also studied the effect of initial fraction of drug (a) on drug release behavior. In case of hydrophilic drugs, the release profile is independent of a (Fig. 4A), [21] which has also been observed for other values of p and w (plots not shown). However, in case of hydrophobic drugs, initial drug loading has substantial effect on the drug release rate (Fig. 4B). This difference in the release behavior is due to the concerted motion of drug molecules when they are part of an aggregate. Higher a values implies higher probability of formation of large aggregates, which results in a progressive reduction in the drug release rate. Finally, we have studied the effect of matrix size on the drug release behavior. As shown in Fig. 5, drug

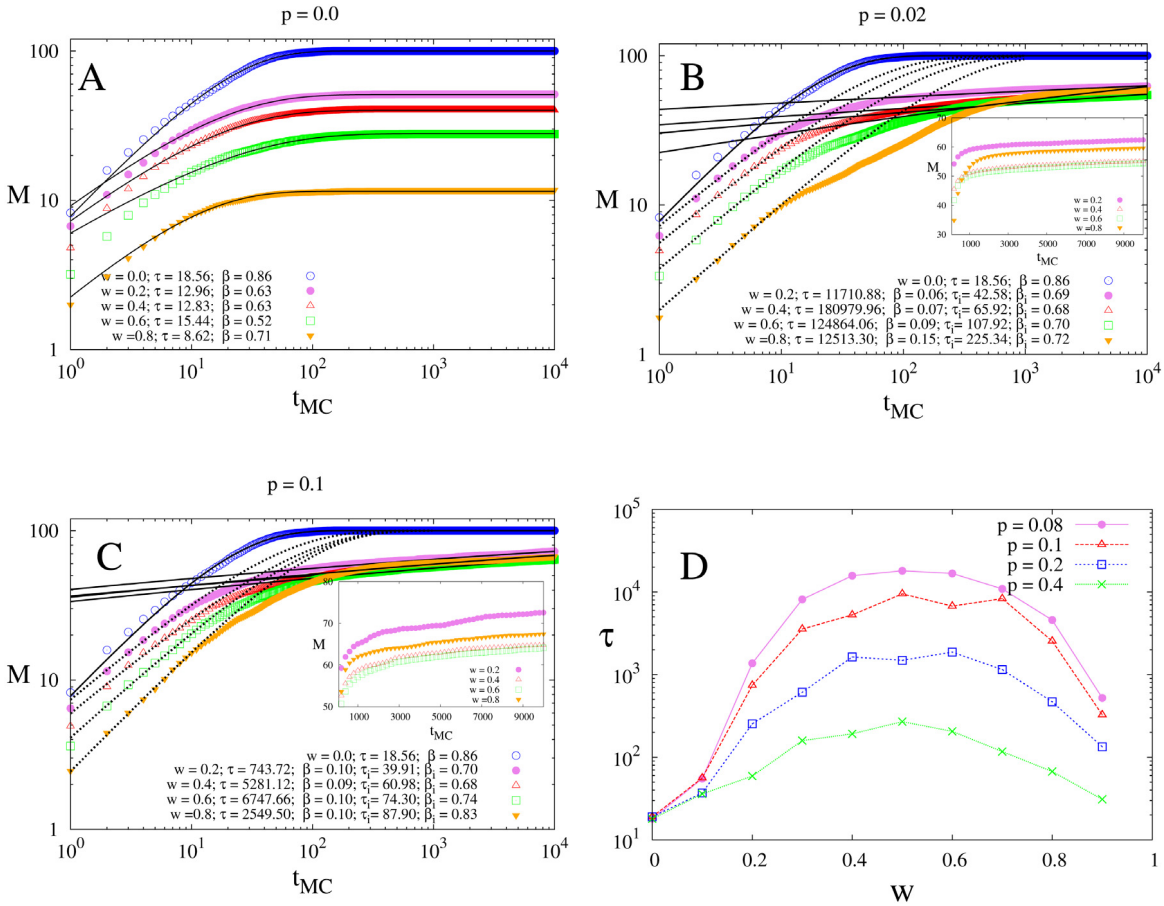

Fig. 3. Friendly ant simulations (hydrophobic drugs). (A), (B) and (C) shows the log–log plot of M versus t_{MC} for $p = 0.0$, $p = 0.02$, and $p = 0.1$, respectively, for $N = 10$, $a = 0.2$, and different values of w. Solid lines ($t_{MC} > 10$) and dotted lines ($t_{MC} \leq 10$) indicate Weibull function fits with the parameters β, τ and β_i, τ_i mentioned in the legends. In (A), M_∞ is different for different w, but in (B) and (C), plots do not saturate till $t_{MC} = 10^4$ but tends towards $M_\infty = 100\%$. Insets in (B) and (C) show the M versus t_{MC} plot for $t_{MC} \geq 100$. (D) shows dependence of τ on w for different p.

release rate decreases with an increase in matrix size in both blind ant (hydrophilic) and friendly ant (hydrophobic) case [47]. Further, the extent of drug release is strongly affected by the matrix size in the friendly ant (hydrophobic) case, indicated by increasingly lower $M(t)$ values for larger matrix size. This happens because the average distance that a drug molecule needs to traverse through the matrix before its escape increases with an increase in the matrix size. In experiments, drug loaded microspheres of different size can be easily prepared [31,32] and therefore matrix size provides an experimentally feasible control on the drug release kinetics. As per our results, smaller matrices will show burst release behavior and larger matrices will show the sustained release behavior, provided the matrix structure and drug distribution is similar. The insets of Fig. 5 show the comparison of $NM_N(t)$ versus t_{MC} for different N, where $M_N(t)$ represents drug release percentage from $N \times N$ matrix. Since $NM_N(t)$ for different N coincide for short-to-intermediate values of time, we can conclude that the $M_N(t)$ is inversely proportional to N [48]. However, substantial deviations are expected at long times, when we approach the saturation limit (M_∞). This regime is more clearly visible in the blind ant case in Fig. 5 than compared to the friendly ant case both until $t_{MC} = 10^4$, since the approach to saturation limit is faster in the blind ant case. The saturation limit M_∞ also decreases in the friendly ant case for $p = 0.0$ (fig. 5C), due to an increased probability of trapping of ant aggregates. Finally, while the aggregates are not trapped for $p = 0.02$ in the friendly ant case (fig. 5D), the approach to saturation is substantially slower than blind ant case and not observed until $t_{MC} = 10^4$.

In order to understand the differences between our approach and conventional approach based on diffusion equation [4,5], we compare the concentration profile of ants obtained with that obtained by the solution of the diffusion equation. For the 2D lattice, we solve the 2D diffusion equation for the drug concentration, $c(x, y, t)$,

$$\frac{\partial c}{\partial t} = D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2}\right), \tag{4}$$

where $c(x, y, 0) = c_0$, c_0 being the initial drug concentration assumed to be uniformly distributed. In Eq. (4), it is assumed that the drug diffusivity D is independent of spatial location. Because of symmetry along x and y directions, it is convenient to

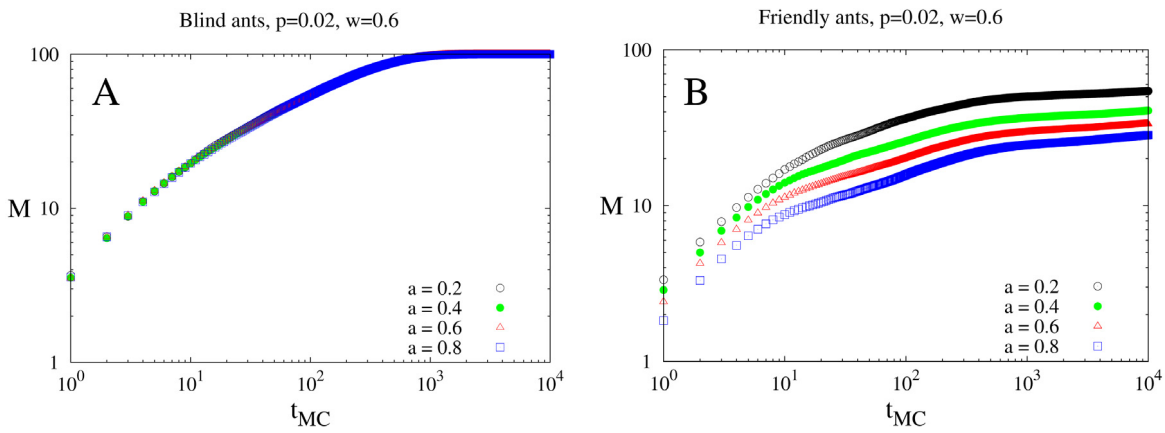

Fig. 4. Effect of initial ant fraction (drug concentration in the dose). Log–log plot of M versus t_{MC} for $a = 0.2, 0.4, 0.6$ and 0.8 in (A) blind ant case for $p = 0.02$, $w = 0.6$ and $N = 10$, (B) friendly ant case for $p = 0.02$, $w = 0.6$ and $N = 10$.

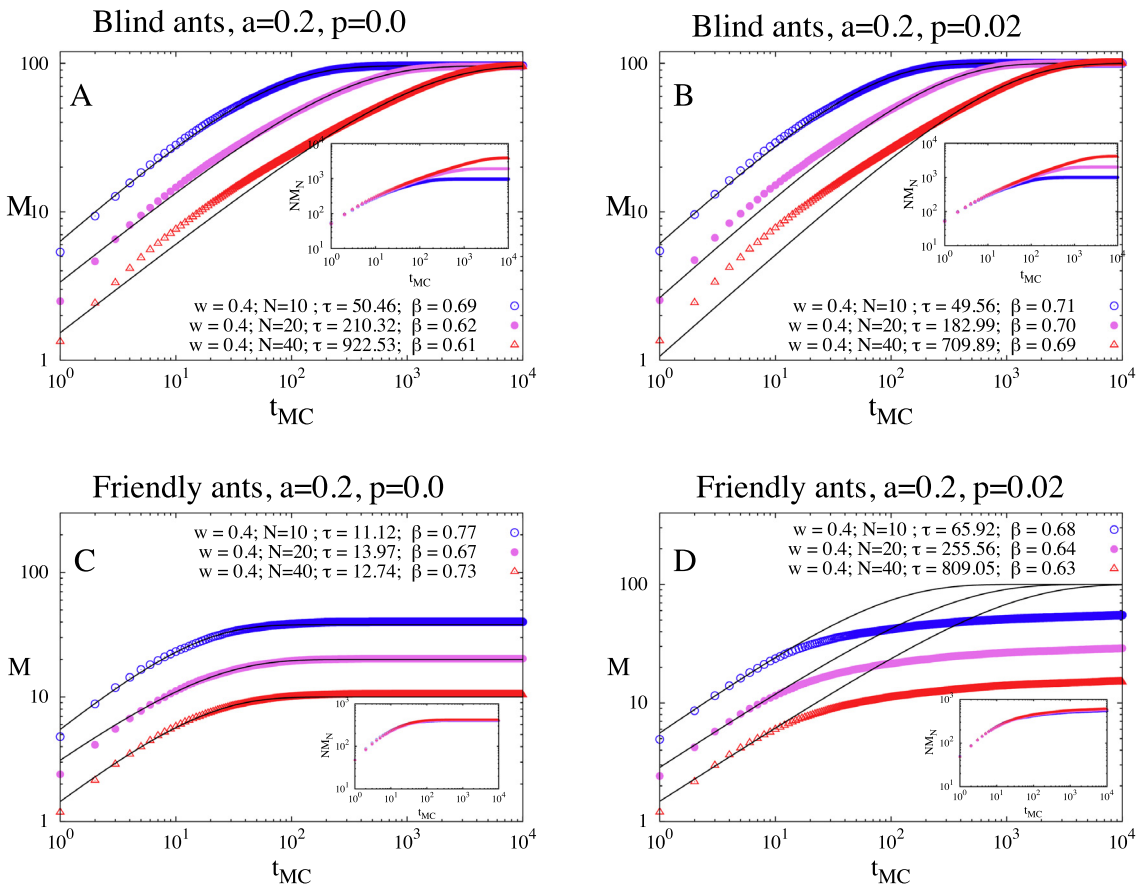

Fig. 5. Effect of matrix size: log–log plot of M vs t_{MC} for $N = 10$, 20 and 40. (A) and (B) compares the matrix size effect on blind ants for $w = 0.4$, $p = 0.0$ and $p = 0.02$ respectively. (C) and (D) compares the matrix size effect on blind ants for $w = 0.4$, $p = 0.0$ and $p = 0.02$ respectively. Inset of each figure shows log–log plot NM_N vs. t_{MC} for $N = 10$, 20 and 40.

use the center of lattice as the origin and solve in the domain $x, y \in (0, L/2)$ with the boundary conditions $\partial c/\partial x = 0$ along $y = 0$ and $\partial c/\partial x = 0$ along $y = 0$. We can use perfect sink conditions $c = 0$ along the open boundaries $x = L/2$ and $y = L/2$. Eq. (4) with the initial and boundary conditions defined above can be solved using variable separation method [5,16] that

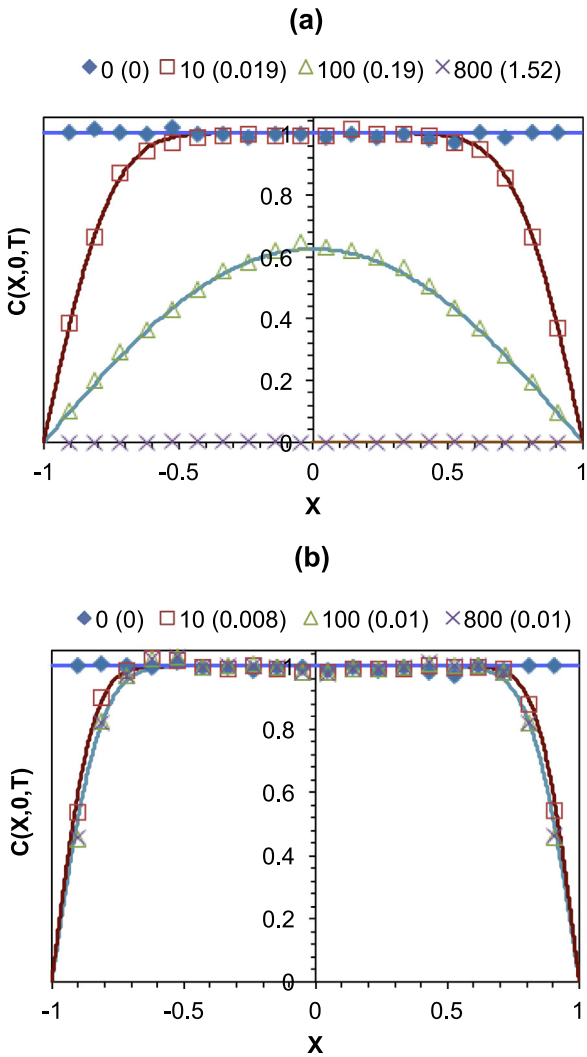

Fig. 6. Plot of dimensionless concentration $C(X, 0, T)$ against dimensionless distance X for (a) blind ants (b) friendly ants. Model parameters are $N = 20$, $a = 0.4$, $p = 0.0$ and $w = 0.1$. Marker represents the simulation data and solid line represents the fit with the equation given in text. The number of MC steps is shown in legends outside the brackets and the dimensionless time used in the fitting equation is given in brackets of the legend.

results in

$$C(X, Y, T) = \frac{16}{\pi^2} \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(-1)^{m+n}}{(2m+1)(2n+1)} \cos\left[(2m+1)\frac{\pi}{2}X\right] \cos\left[(2n+1)\frac{\pi}{2}Y\right] \exp\left[-\left\{(2m+1)^2 + (2n+1)^2\right\}\frac{\pi^2}{4}T\right], \tag{5}$$

where we have introduced dimensionless variables, $C = c/c_0$, $X = x/(L/2)$, $Y = y/(L/2)$, and $T = t/\tau_0$ for $\tau_0 = L^2/4D$.

In Fig. 6, we show the representative profiles of $C(X, 0, T)$ for blind ant (non-aggregating) and friendly ant (aggregating) cases, where we have used symmetry $C(-X, 0, T) = C(X, 0, T)$ to get the plot for $X \in (-1, 1)$. At MC steps $= 0$, ants are distributed uniformly inside the matrix for both cases. Initially, for both the cases, the ants near the boundary escapes and the density of ants deep inside the matrix does not change significantly (see MC steps$=$ 10 data in Fig. 6). Afterwards, for blind ant case, due to concentration gradient, ants in the middle of the matrix diffuses outward and concentration decreases everywhere (MC steps $= 100$), and after sufficiently long time (MC steps$=$ 800), all ants escape. But for friendly ant case, due to the formation of aggregates and the presence of walls, most of the ants got trapped and density profile does not change after MC steps $= 10$. From data fitting, we found that MC steps and T are related to each other by $T = k \times$(MC steps), where k is a proportionality constant and its value do not change with time for blind ants ($k = 0.0019$) but for friendly ants k depends on time. We have also checked and found that k value is constant for both type of ants in absence of any walls. That

is, D is constant in absence of walls. However, constant D is observed only for low wall fraction for the blind ant case and subdiffusive behavior is obtained for larger wall fractions (see $w = 0.4$ case in Figure 1 of supplementary information). This implies that the real time $t \propto$ MC steps in the blind ant case for low wall fractions because both L and D are constant and hence $k\tau_0 = kL^2/4D$ is constant. But, in blind ant case at high wall fractions and friendly ant case, D varies with time, which means that τ_0 is not constant and thus t is not proportional to MC steps.

4. Conclusions

Monte Carlo simulations performed on the "ant-wall" model developed in this work captures the effects of matrix structure, drug loading, drug hydrophobicity, excipient–drug interaction, and matrix size within the simple framework of a 2D lattice model. Following list summarizes the key findings of our simulations:

1. Drug release profile of hydrophilic drugs is reliably fit using a Weibull function in the entire time range with minor deviations occurring in the short-time behavior for high drug loading. Drug release profile of hydrophobic drugs can be explained by two Weibull functions for the initial burst release and the long-time sustained release, respectively.
2. Significant trapping of drug molecules occur in the case of strong excipient–drug interactions thus resulting in an incomplete drug release ($M_\infty < 100\%$). On the other hand, excipients provide little control over the drug release behavior for weak excipient–drug interactions. Moderate excipient–drug interactions ($\Delta E \sim 3 - 5k_BT$) are therefore ideal for sustained drug release [49,50].
3. In the case of hydrophobic drugs, there is an optimal value of drug loading near the percolation threshold of excipient, which provides ideal condition for sustained drug release. This arises from a fine interplay between aggregation kinetics and drug escape in the case of hydrophobic drugs.
4. Drug release profile is independent of the drug concentration in the dose for a given value of drug loading in the case of hydrophilic drugs. But, substantial effects of dose concentration of the drug on the drug release profile is observed in the hydrophobic case, where the rate and extent of drug release decreases with increase in dose concentration.
5. Drug release is inversely proportional to matrix size(N) [51] for both hydrophilic and hydrophobic drugs. However, even the extent of drug release is strongly affected by the matrix size in the case of hydrophobic drugs. The qualitative profile of drug release is independent of matrix size.

Several extensions of this study are possible and worth pursuing. We have only considered two extreme cases of highly hydrophilic (blind ant) and highly hydrophobic drugs (friendly ant). Introduction of a parameter related to the degree of hydrophobicity will be needed to extend the model to intermediate cases. Further, effects of matrix erosion and swelling-induced structural changes can be included, by introduction of a mechanism to manipulate matrix structure during simulations. Finally, simulations can be performed for a range of matrix structures and lattice dimensions, in order to study different types of matrix formulations.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.physa.2018.12.029.

References

[1] A. Serajuddin, Solid dispersion of poorly water-soluble drugs: Early promises, subsequent problems, and recent breakthroughs, J. Pharmaceutical Sci. 88 (10) (1999) 1058–1066.
[2] D.Q. Craig, The mechanisms of drug release from solid dispersions in water-soluble polymers, Int. J. Pharmaceutics 231 (2) (2002) 131–144.
[3] C. Maderuelo, A. Zarzuelo, J.M. Lanao, Critical factors in the release of drugs from sustained release hydrophilic matrices, J. Controlled Release 154 (1) (2011) 2–19.
[4] J. Siepmann, F. Siepmann, Mathematical modeling of drug delivery, Int. J. Pharmaceutics 364 (2) (2008) 328–343.
[5] J. Siepmann, F. Siepmann, Modeling of diffusion controlled drug delivery, J. Control. Release 161 (2) (2012) 351–362.
[6] S.-F. Chou, D. Carson, K.A. Woodrow, Current strategies for sustaining drug release from electrospun nanofibers, J. Control. Release 220 (Part B) (2015) 584–591.
[7] I. Sebe, P. Szabo, B. Kallai-Szaba, R. Zelko, Incorporating small molecules or biologics into nanofibers for optimized drug release: A review, Int. J. Pharmaceutics 494 (1) (2015) 516–530.
[8] A. Salehi, J. Zhao, T.D. Cabelka, R.G. Larson, A unified multicomponent stress-diffusion model of drug release from non-biodegradable polymeric matrix tablets, J. Control. Release 224 (Supplement C) (2016) 43–58.
[9] R.W. Korsmeyer, R. Gurny, E. Doelker, P. Buri, N.A. Peppas, Mechanisms of solute release from porous hydrophilic polymers, Int. J. Pharmaceutics 15 (1) (1983) 25–35.
[10] W. Weibull, A statistical distribution function of wide applicability, J. Appl. Mech. 18 (1951) 293–297.
[11] S. Casault, G.W. Slater, Systematic characterization of drug release profiles from finite-sized hydrogels, Physica A 387 (22) (2008) 5387–5402.

[12] S. Casault, G.W. Slater, Comments concerning: Monte Carlo simulations for the study of drug release from matrices with high and low diffusivity areas, Int. J. Pharmaceutics 365 (1) (2009) 214–215.
[13] H.B. Hopfenberg, Controlled release from erodible slabs, cylinders, and spheres, in: Controlled Release Polymeric Formulations, 1976, pp. 26–32, (Chapter).
[14] M. Ignacio, M.V. Chubynsky, G.W. Slater, Interpreting the Weibull fitting parameters for diffusion-controlled release data, Physica A 486 (2017) 486–496.
[15] T. Higuchi, Rate of release of medicaments from ointment bases containing drugs in suspension, J. Pharm. Sci. 50 (10) (1961) 874–875.
[16] R.D. Manga, P.K. Jha, Mathematical models for controlled drug release through ph-responsive polymeric hydrogels, J. Pharm. Sci. 106 (2) (2017) 629–638.
[17] A. Bunde, S. Havlin, R. Nossal, H.E. Stanley, G.H. Weiss, On controlled diffusion-limited drug release from a leaky matrix, J. Chem. Phys. 83 (11) (1985) 5909–5913.
[18] A. Balazs, D. Calef, J. Deutch, R. Siegel, R. Langer, The role of polymer matrix structure and interparticle interactions in diffusion-limited drug release, Biophys. J. 47 (1) (1985) 97–104.
[19] R.A. Siegel, R. Langer, Mechanistic studies of macromolecular drug release from macroporous polymers. II. Models for the slow kinetics of drug release, J. Control. Release 14 (2) (1990) 153–167.
[20] K. Kosmidis, P. Argyrakis, P. Macheras, A reappraisal of drug release laws using monte carlo simulations: the prevalence of the weibull function, Pharmaceutical Res. 20 (7) (2003) 988–995.
[21] K. Kosmidis, P. Argyrakis, P. Macheras, Fractal kinetics in drug release from finite fractal matrices, J. Chem. Phys. 119 (12) (2003) 6373–6377.
[22] A. Barat, H.J. Ruskin, M. Crane, Probabilistic models for drug dissolution. Part 1. Review of Monte Carlo and stochastic cellular automata approaches, Simul. Model. Pract. Theory 14 (7) (2006) 843–856.
[23] A. Barat, H.J. Ruskin, M. Crane, Probabilistic methods for drug dissolution. Part 2. Modelling a soluble binary drug delivery system dissolving in vitro, Simul. Model. Pract. Theory 14 (7) (2006) 857–873.
[24] R. Villalobos, S. Cordero, A.M. Vidales, A. Domínguez, In silico study on the effects of matrix structure in controlled drug release, Physica A 367 (2006) 305–318.
[25] K. Kosmidis, P. Macheras, Monte Carlo simulations for the study of drug release from matrices with high and low diffusivity areas, Int. J. Pharmaceutics 343 (1) (2007) 166–172.
[26] L. Martínez, R. Villalobos, M. Sánchez, J. Cruz, A. Ganem, L.M. Melgoza, Monte Carlo simulations for the study of drug release from cylindrical matrix systems with an inert nucleus, Int. J. Pharmaceutics 369 (1) (2009) 38–46.
[27] A. Hadjitheodorou, G. Kalosakas, Quantifying diffusion-controlled drug release from spherical devices using Monte Carlo simulations, Mater. Sci. Eng. C 33 (2) (2013) 763–768.
[28] A. Hadjitheodorou, G. Kalosakas, Analytical and numerical study of diffusion-controlled drug release from composite spherical matrices, Mater. Sci. Eng. C 42 (2014) 681–690.
[29] M.S.G. Filho, F.A. Oliveira, M.A.A. Barbosa, A statistical mechanical model for drug release: Investigations on size and porosity dependence, Physica A 460 (2016) 29–37.
[30] J. Siepmann, N. Peppas, Modeling of drug release from delivery systems based on hydroxypropyl methylcellulose (HPMC), Adv. Drug Deliv. Rev. 64 (2012) 163–174.
[31] C. Berkland, M. King, A. Cox, K.K. Kim, D.W. Pack, Precise control of PLG microsphere size provides enhanced control of drug release rate, J. Control. Release 82 (1) (2002) 137–147.
[32] C. Berkland, M.J. Kipper, B. Narasimhan, K.K. Kim, D.W. Pack, Microsphere size, precipitation kinetics and drug distribution control drug release from biodegradable polyanhydride microspheres, J. Control. Release 94 (1) (2004) 129–141.
[33] H. Leuenberger, J. Bonny, M. Kolb, Percolation effects in matrix-type controlled drug release systems, Int. J. Pharmaceutics 115 (2) (1995) 217–224.
[34] R. Pandey, D. Stauffer, Fractal dimensionality and the number of visited sites of the ant in the labyrinth, J. Phys. A: Math. Gen. 16 (14) (1983) L511.
[35] J.P. Straley, The ant in the labyrinth: diffusion in random networks near the percolation threshold, J. Phys. C: Solid State Phys. 13 (16) (1980) 2991.
[36] I. Majid, D.B. Avraham, S. Havlin, H.E. Stanley, Exact-enumeration approach to random walks on percolation clusters in two dimensions, Phys. Rev. B 30 (1984) 1626–1628.
[37] J.-P. Bouchaud, A. Georges, Anomalous diffusion in disordered media: Statistical mechanisms, models and physical applications, Phys. Rep. 195 (4) (1990) 127–293.
[38] S. Havlin, D. Ben-Avraham, Diffusion in disordered media, Adv. Phys. 36 (6) (1987) 695–798.
[39] J. Haus, K. Kehr, Diffusion in regular and disordered lattices, Phys. Rep. 150 (5) (1987) 263–406.
[40] P.K. Jha, V. Kuzovkov, B.A. Grzybowski, M.O. de la Cruz, Dynamic self-assembly of photo-switchable nanoparticles, Soft Matter 8 (1) (2012) 227–234.
[41] R. Zhang, P.K. Jha, M.O. de la Cruz, Non-equilibrium ionic assemblies of oppositely charged nanoparticles, Soft Matter 9 (20) (2013) 5042–5051.
[42] C.L. Perrin, J.B. Nielson, "Strong" hydrogen bonds in chemistry nd biology, Annual Review of Physical Chemistry 48 (1) (1997) 511–544.
[43] M. Cheng, Z. Qin, S. Hu, S. Dong, Z. Ren, H. Yu, Achieving long-term sustained drug delivery for electrospun biopolyester nanofibrous membranes by introducing cellulose nanocrystals, ACS Biomater. Sci. Eng. 3 (8) (2017) 1666–1676.
[44] S.T. Yohe, Y.L. Colson, M.W. Grinstaff, Superhydrophobic materials for tunable drug release: using displacement of air to control delivery rates, J. Am. Chem. Soc. 134 (4) (2012) 2016–2019, PMID: 22279966.
[45] R. Kopelman, Fractal reaction kinetics, Science 241 (4873) (1988) 1620–1626.
[46] C. Santanu, K. Madhusmruti, S. Anuradha, C. Patra, P. V., S. Kalyan, Effects of drug solubility on the release kinetics of water soluble and insoluble drugs from HPMC based matrix formulations, Vol. 59, 2009, p. 313.
[47] G. Kalosakas, D. Martini, Drug release from slabs and the effects of surface roughness, Int. J. Pharmaceutics 496 (2) (2015) 291–298.
[48] P.L. Ritger, N.A. Peppas, A simple equation for description of solute release I. Fickian and non-fickian release from non-swellable devices in the form of slabs, spheres, cylinders or discs, J. Control. Release 5 (1) (1987) 23–36.
[49] S. Puttipipatkhachorn, J. Nunthanid, K. Yamamoto, G. Peck, Drug physical state and drug-polymer interaction on drug release from chitosan matrix films, J. Control. Release 75 (1) (2001) 143–153.
[50] S. Baghel, H. Cathcart, N.J. O'Reilly, Theoretical and experimental investigation of drug-polymer interaction and miscibility and its impact on drug supersaturation in aqueous medium, Eur. J. Pharmaceut. Biopharmaceut. 107 (2016) 16–31.
[51] J. Siepmann, H. Kranz, N. Peppas, R. Bodmeier, Calculation of the required size and shape of hydroxypropyl methylcellulose matrices to achieve desired drug release profiles, Int. J. Pharmaceutics 201 (2) (2000) 151–164.